

Predicting Hit Songs Using Repeated Chorus

Dhanush Sai Suprapadha - PES2UG24CS154
Deepthi V - PES2UG24CS150
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Problem and dataset

We test whether 15-second chorus features can distinguish stronger chart success, following Eric Liu's CS229 project. The experiment uses a public Apache-2.0 dataset of **751 songs from 71 artists**, with 518 numerical audio features. There are 366 year-end-hit labels and 385 other-chart-song labels. This public dataset defines a hit proxy different from the paper's charted/uncharted target. Its original audio is unavailable.

Method and implementation

Eleven audio feature families produce 74 channels, each summarized by seven statistics. A fixed split holds out 154 songs from 19 artists, leaving 597 songs from 52 artists for training. Five artist-disjoint training folds select model settings by balanced accuracy. Preprocessing fits inside each fold. PCA keeps 95% of training variance for non-tree models. We compare logistic regression, LDA, three SVM kernels, random forest, gradient boosting, a neural network, and a majority-class baseline.

Measured results

Metric	Selected polynomial SVM	Majority baseline
Training CV balanced accuracy	53.6%	50.0%
Test balanced accuracy	46.6%	50.0%
Test accuracy	46.8%	50.6%
Test F1 / ROC-AUC	0.406 / 0.456	0.000 / 0.500

The 95% interval for selected-model balanced accuracy is 38.7%-55.8%, using 2,000 bootstrap resamples of test artists. The selected model stays unchanged after test evaluation.

Conclusion and demonstration

The experiment finds **no reliable predictive advantage over the baseline for unfamiliar artists**. This is an empirical limitation of the current data and features, not proof about all music. Historical sampling, inherited proxy labels, and unavailable original recordings limit interpretation. Future work should verify labels and consistently re-extract licensed audio before further modeling.

The working Streamlit demo predicts held-out songs, displays model comparisons and errors, and supports exploratory audio uploads with automatic or manual 15-second selection. The repository includes the code, notebook, dependency versions, trained model, data provenance, split manifest, and reproducible results.

Sources

[Eric Liu, CS229 report \(2021\)](#). [Antonios Malak, dataset and notebooks](#) (Apache-2.0, commit 838e76f). Full provenance and method details appear in the repository.